

Rohit Yadav

Built 25+ LLMs from scratch • 120M–1.4B params • TPU clusters • Trained on 500 Billion+ tokens

+91 8849994303 | yrohit1825@gmail.com | linkedin.com/in/rohit-yadav-25535b256 | github.com/YADAV1825 | rohit-portfolio-yadav1825.vercel.app

EDUCATION

National Institute of Technology, Jalandhar

Bachelor of Technology in Information Technology

Green Valley School for Children

CBSE Board 12th

Jalandhar, India

Aug 2023 – Aug 2027

Gandhinagar, India

May 2021 – May 2023

PROFESSIONAL SUMMARY

Third-year undergraduate and Systems + AI builder who enjoys building complex systems from scratch. Built 25+ LLMs from scratch (120M–1.4B parameters) on Google's TPU clusters, working with 10TB+ datasets (PILE, DOLMA) and 500B+ tokens. Experienced in **pretraining, fine-tuning, and continual pretraining** of large-scale models. Designed high-throughput data pipelines and tokenization workflows for distributed training. Experience developing clinical-scale genomic ML systems and performance-critical infrastructure. Awarded Google TPU Research Cloud (TRC) grant and AMD MI300X compute access for large-scale experimentation.

EXPERIENCE

Full-Stack Web Developer Intern

Skylark Express Delhi Pvt Ltd

Jun 2025 – Jul 2025

On-site, Gurugram

- Contributed to the development of an internal logistics tracking dashboard, improving data visibility and operational workflows.
- Built and optimized frontend components using **React, Next.js, and TypeScript**, focusing on performance and reusability.
- Integrated **REST APIs** to enable seamless data flow between backend services and user interfaces.
- Improved UI responsiveness and system efficiency, enhancing overall user experience for internal teams.

PROJECTS

AutonomousX: TPU-Native LLM Training Organization (Instinct Model Family)

[\[HuggingFace\]](#)

- Built and open-sourced **25+ LLMs (120M–1.4B params)** by engineering **TPU-native distributed pipelines (JAX pmap)** under AutonomousX.
- Trained the **Instinct model family** on **500B+ tokens** (PILE, DOLMA) under extreme overtraining regimes to study **scaling laws, convergence, and degeneration dynamics**.
- Implemented efficient dataset streaming, binary tokenization (.bin), and multi-host checkpointing on **Google TPU pods** without any external team support.
- Open-sourced complete pipelines—including HuggingFace weights, training logs, and a 10-minute **TPU setup guide** enabling full reproducibility for researchers.

PathoPreter: Clinical-Grade SNV Pathogenicity Ranker (DNA Foundation Model)

[\[HuggingFace\]](#)

- Developed a **hybrid genomic foundation model (500M params)** fusing a Nucleotide Transformer backbone with clinical tabular features for **SNV pathogenicity prediction**.
- Achieved **0.9186 ROC-AUC** on a rigorous 14k rare-variant benchmark, outperforming industry SOTA (AlphaMissense, CADD) by **+0.32 ROC-AUC**.
- Designed a **ranking-first clinical triage system** demonstrating **75%+ pathogen recall** while testing only 5–10% of variants, drastically reducing lab turnaround time and costs.
- Validated biological learning via **SHAP ablation studies** (showing 64.9% reliance on raw DNA) and open-sourced the **end-to-end reproducible pipeline**.

PathoPreter V2: Explainable ML + LLM Pipeline for Clinical SNV Interpretation

[\[GitHub\]](#)

- Built **PathoScan**, a high-performance clinical interpretation system combining an **XGBoost classifier** with **LLM-based reasoning** over SHAP visualizations.

- Achieved **0.97+ ROC-AUC** on large real-world datasets and **0.767 ROC-AUC** on a 14k hard benchmark, outperforming established tools like CADD and REVEL.
- Engineered strict **leakage-free datasets** and validated robustness via permutation tests, showing dominant learning reliance on evolutionary conservation over positional bias.
- Deployed an **interactive web platform** optimized for clinical decision-making, featuring real-time predictions and automated **LLM-generated clinical insight reports**.

Large-Scale ClinVar SNV Dataset Pipeline for Genomic ML (1.25M Variants) [\[Dataset Pipeline\]](#)

- Architected an **end-to-end 32-core parallelized data pipeline**, reducing processing time of **250GB+ genomic data** from 150 hours to under 20 hours.
- Transformed **7.7M raw ClinVar records** into a pristine, leakage-free dataset of **1.25M high-quality SNVs** optimized for ML evaluation and benchmarking.
- Integrated multi-source clinical data (gnomAD, dbNSFP, hg38) to generate **50+ predictive features** alongside raw GRCh38 DNA sequence mapping.
- Enforced strict preprocessing to eliminate HGVS identifiers, providing a balanced **300K subset** tailored for stable, reliable transformer-based model training.

BroLang Compiler + VM Integration [\[GitHub\]](#)

- Designed a complete compiler from scratch for a custom language (BroLang), implementing a **Lexer, Parser, AST, and Bytecode Generator**.
- Seamlessly integrated the compiler output with a **custom 16-bit Virtual Machine** for direct bytecode execution on virtual hardware.
- Engineered a self-contained execution ecosystem fully supporting complex conditional statements, loops, and standard I/O print functions.
- **Tech Stack:** Modern C++17, custom binary bytecode formatting, and advanced compiler pipeline design principles.

RohitVM + Virtual CPU (16-bit Architecture) [\[GitHub\]](#)

- Developed a highly optimized **16-bit CPU emulator** featuring a fully functional 65KB RAM, hardware registers, and dynamic stack memory.
- Implemented a robust **custom instruction set architecture (ISA)** including core operations: MOV, ADD, SUB, JMP, CMP, PUSH/POP, PRINT, and HALT.
- Built the complete runtime execution environment designed to natively decode, track, and execute bytecode originating from the BroLang compiler.
- **Tech Stack:** C++, explicit manual memory management, instruction decoding algorithms, and high-performance runtime loops.

TECHNICAL SKILLS

Languages: C++, Python, TypeScript/JavaScript, SQL

AI & Distributed Training: LLM Pretraining & Fine-tuning, TPU (v4/v6e) & GPU (A100/H100) Clusters, JAX pmap, Scaling Laws, Mixed Precision (BF16), Gradient Checkpointing, XAI (SHAP)

Frameworks & Web: JAX/Flax, PyTorch, HuggingFace, React, Next.js, FastAPI, PostgreSQL, MongoDB

Cloud & Systems Architecture: GCP (TPU Research Cloud), AWS, Docker, Linux, Compiler Design, Virtual Machines, POSIX Sockets

ACHIEVEMENTS

Awarded Google TRC grant for 320 TPU access (SPOT + STANDARD VMs).

Awarded AMD MI300X GPU access for 300 non-preemptive hours

IOQM (PRMO): Qualified for Indian Olympiad Qualifier in Mathematics

KVPY SA: AIR 2590

LeetCode: Knight (2013), 300+ problems solved

CodeChef: 4-Star (1818)

Codeforces: Specialist (1417)

CodeChef START166B: Global Rank 9

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